

Course Introduction

Welcome to Python Programming

Control flow and conditionals

✔ **Video:** Math and logical operators
7 min

✔ **Video:** Control flow: If / else, else if
6 min

✔ **Reading:** Conditional statements
15 min

✔ **Video:** Switch statement
6 min

✔ **Video:** Looping constructs
5 min

✔ **Reading:** Looping Constructs: Practical Examples
5 min

📖 **Reading:** Practicing control flow and loops
5 min

📺 **Video:** Nested loops and the effect on algorithmic complexity
5 min

📖 **Reading:** Exercise: Use control flow and loops to solve a problem
30 min

📖 **Reading:** Use control flow and loops to solve a problem - solution
10 min

📖 **Practice Quiz:** Self-review: Use control flow and loops to solve a problem
3 questions

📺 **Video:** Module summary: Getting started with Python
1 min

📖 **Quiz:** Module quiz: Getting started with Python
10 questions

📖 **Reading:** Additional resources
5 min

Looping Constructs: Practical Examples

This reading introduces you to the different looping constructs in Python.

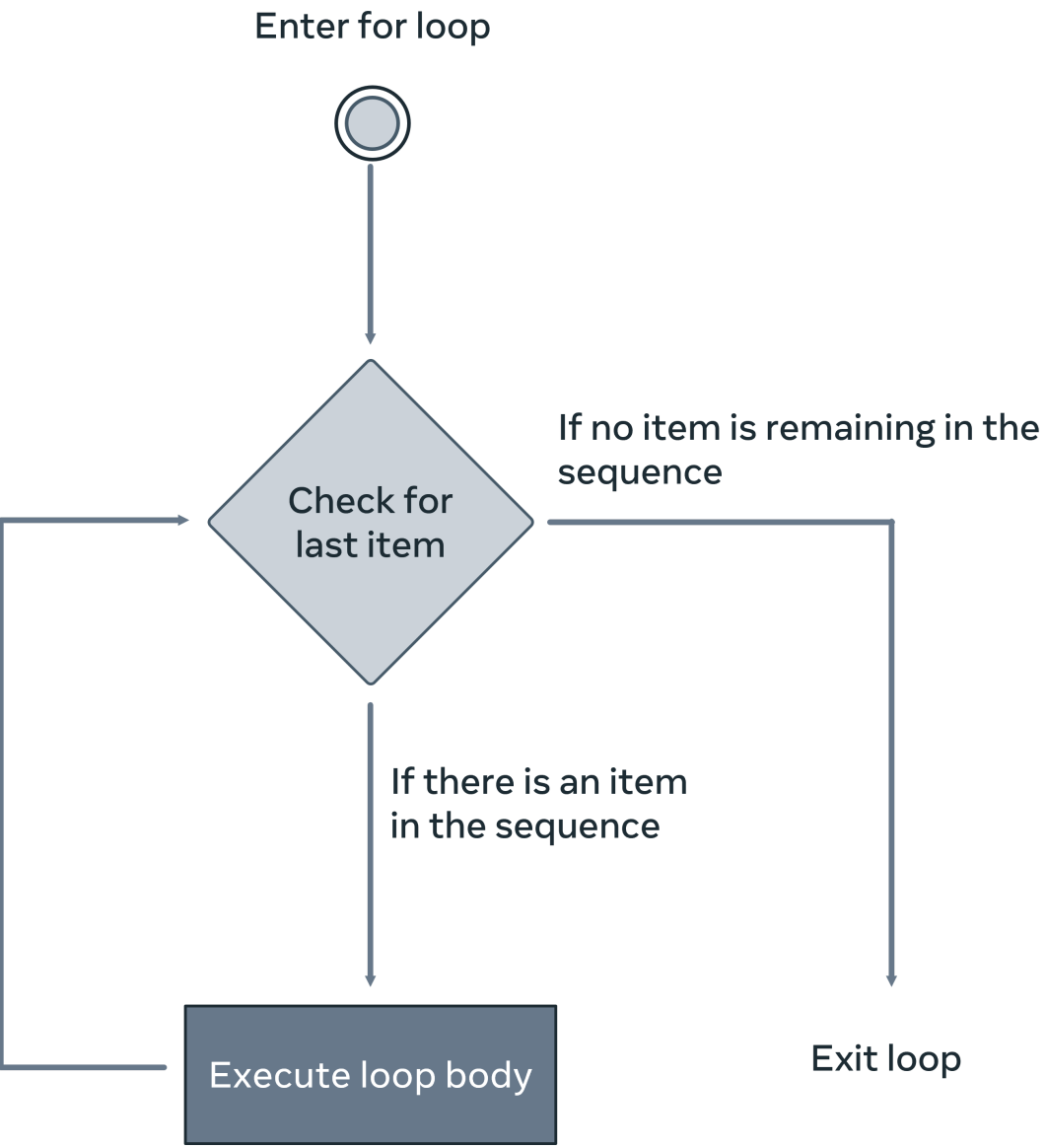
For loop

Looping through data is a fairly common task in any programming language. The `for` loop makes it easy to work with any type of sequence in Python. Let's run through some examples of `for` loops and the different ways you can use them.

```
1 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
2
3 for dessert in favorites:
4     print('One of my favorite desserts is', dessert)
5
6
```

In the code snippet above, the `for` loop iterates over the contents of the `favorites` list and prints out a sentence with the dessert name for each item in the list.

The `for` loop is based on the size or length of the elements to iterate over.



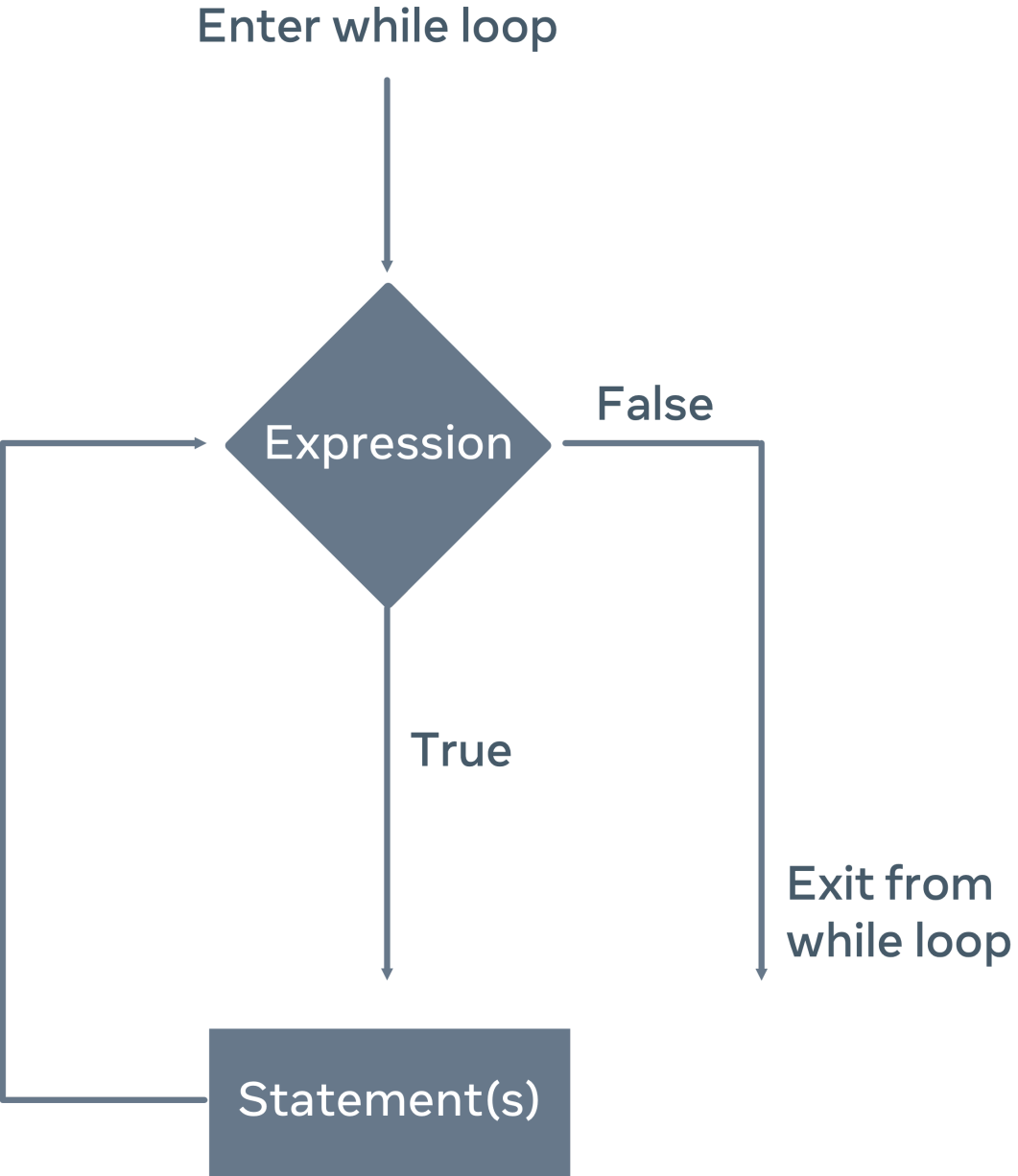
While loop

On the other hand, a `while` loop is based upon a condition being true. Once the condition is no longer true the loop stops. The amount of times the `while` loop is executed is not known ahead of time as it is with the `for` loop.

If you take the above `for` loop example and convert that to the `while` loop alternative, you will end up with something like this:

```
1 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
2
3 count = 0
4
5 while count < len(favorites):
6     print('One of my favorite desserts is', favorites[count]);
7     count += 1
8
```

Note that you needed to declare a `count` variable. The `count` variable is then compared to the length of the `favorites` list. As you loop through the data the `count` is incremented. Once the condition `count < len(favorites)` is no longer true the loop will stop.



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